



Shahadat Hussain

Doctor of Philosophy · Research Associate at New York University Abu Dhabi

3D Printing / Additive Manufacturing, Composite filaments, TPMS, Metamaterials, Materials Science, & Flexural Testing

About

Publications (18)

Questions (1)

Network

Contact

About

18Publications

6,988Reads

435Citations

or

Introduction

Dr. Shahadat Hussain is a Mechanical Engineering researcher specializing in 3D printing of carbon fiber–reinforced polymer composites and metamaterial structures, including cubes and beams. His experimental research focuses on flexural and compression testing, mechanical characterization, and structural performance of additively manufactured composite materials for advanced engineering applications.

Skills and Expertise

MicrostructureHeat TreatmentMechanical PropertiesMaterial CharacterizationX-ray DiffractionTensile TestingSEM AnalysisMaterials ProcessingMechanical TestingAdditive Manufacturing

Additional affiliations

February 2020 - June 2024

Khalifa University

Department of Mechanical Engineering · India

Position

PostDoc

Description

Additive manufacturing of NITI shape memory alloys



August 2014 - February 2017

Advanced Materials and Processes Research Institute

Smart and Functional Materials · Bhopal, India

Position

PhD Student



September 2012 - May 2013

Oriental college of technology

Mechanical Engineering · India

Position

Lecturer

Description

1.Machine Component Design 2. Strength of Material 3.Material Science 4.Basic Mechanical Engineering



Show All

Education

July 2014 - June 2019

Academy of Scientific and Innovative Research

Field of study

Materials Science and Technology



August 2008 - June 2012

Rajiv Gandhi Technical University

Field of study

Mechanical Engineering



Publications

Publications (18)

Phase Transformation Behavior of NITI Triply Periodic Minimal Surface Lattices Fabricated by Laser Powder Bed Fusion

ArticleFeb 2024

Shahadat Hussain · Ali Alagha · Wael Zaki

Laser powder bed fusion (LPBF) is an appealing additive manufacturing technique suitable for producing intricate products with complex shapes, where traditional subtractive manufacturing methods may be impractical. When dealing with NITI shape memory alloys, the fabrication process encounters additional challenges due to the material's high ductil...

View

Microstructural and surface analysis of NITI TPMS lattice sections fabricated by laser powder bed fusion

ArticleSep 2023

Shahadat Hussain · Ali Alagha · Gregory N Haidemenopoulos · Wael Zaki

Due to their high ductility and superior strength, the machining of NITI shape memory alloys using conventional subtractive manufacturing technologies poses significant challenges. However, additive manufacturing (AM) offers a viable solution by circumventing the limitations of machinability through the elimination of tooling, thereby enabling the...

View

Inhomogeneous Microstructure due to Non-Uniform Solidification Rate in NITI Triply Periodic Minimal Surface (TPMS) Structures Fabricated via Laser Powder Bed Fusion

Conference PaperFull-text availableFeb 2023

Shahadat Hussain · Ali Alagha · Wael Zaki

In recent times, interest in the fabrication of porous NITI structures have grown significantly. Porous structures have remarkable potential to be used in the areas of tissue engineering, impact absorption, and fluid permeability. However, fabrication of NITI structures poses challenges such as poor machinability, high work hardening, and inherent...

View

Phase Transformation Behavior of Niti Triply Periodic Minimal Surface Lattices Fabricated by Laser Powder Bed Fusion

PreprintJan 2023

Shahadat Hussain · Ali Alagha · Gregory N Haidemenopoulos · Wael Zaki

View

Imperfections Formation in Thin Layers of NITI Triply Periodic Minimal Surface Lattices Fabricated Using Laser Powder Bed Fusion

ArticleFull-text availableNov 2022

Shahadat Hussain · Ali Alagha · Wael Zaki

In this paper, thin layers of NITI shape memory alloy (SMA) triply periodic minimal surface lattices (TPMS) are fabricated using laser powder bed fusion (LPBF), considering different laser scanning strategies and relative densities. The obtained architected samples are studied using experimental methods to characterize their microstructural feature...

View

Additive manufacturing of shape memory alloys: A review with emphasis on powder bed systems

ArticleFull-text availableMar 2021

Ali Alagha · Shahadat Hussain · Wael Zaki

The ability of shape memory alloys (SMAs) to recover inelastic strains larger than any other metallic alloy has prompted their use in a wide range of applications. However, for the most common SMAs, including NITI and Cu-based systems, fabrication using conventional means raises important challenges, including poor workability and potentially high...

View

Influence of niobium and silver on mechanical properties and shape memory behavior of Cu-12Al-4Mn alloys

ArticleMay 2020

Abhishek Pandey · Shahadat Hussain · prasanth Nair · Rupa Dasgupta

View

Nano-CeO2 Doped Cu–Al–Ni SMAs with Enhanced Mechanical as well as Shape Recovery Characteristics

ArticleDec 2019

Shahadat Hussain · Abhishek Pandey · Rupa Dasgupta

Effect of nano-CeO2 doping was studied on self-accommodating Cu–12Al–4Ni based shape memory alloys. Cu–Al–Ni based alloys are known for their high thermal stability and good shape memory properties but mechanical processing of these alloys is very difficult due to their high brittle nature and susceptibility to inter-granular failure. The present w...

View

Designed polycrystalline ultra-high ductile boron doped Cu-Al-Ni based shape memory alloy

ArticleDec 2018

Shahadat Hussain · Abhishek Pandey · Rupa Dasgupta

Cu–Al–Ni shape memory alloys have good thermal stability and shape memory properties, but mechanical working of these alloys is very difficult due to inter-granular fracture and brittleness. This paper deals with improvement of shape memory properties and ductility of these alloys with the addition of boron. It was observed that with addition of bo...

View

Effect of Alloying Additions on the Properties Affecting Shape Memory Properties of Cu–12.5Al–5Mn Alloy

ChapterJan 2018

Rupa Dasgupta · Ashish Kumar Jain · Shahadat Hussain · [...] · Sampath Vedamanickam

The paper discusses the attempt made to understand the role of different alloying additions (Fe, Ni, Cr and Ti) to a Cu–12.5Al–5Mn alloy known to exhibit shape memory behaviour on its microstructure, phase precipitation and effect on transition temperatures after optimising the heat treatment cycles to precipitate the required martensite phase. The...

View

Study of effect of Fe, Cr and Ti on the martensite phase formation in Cu-12.5wt%Al-5wt%Mn SMA

ArticleDec 2016

Shahadat Hussain · Ashish K. Jain · Mohammad Ayub Ansari · [...] · Rupa Dasgupta

View

Effect of Nano CeO2 Addition on the Microstructure and Properties of a Cu-Al-Ni Shape Memory Alloy

ArticleMay 2016

Abhishek Pandey · Ashish Kumar Jain · Shahadat Hussain · [...] · Rupa Dasgupta

This article deals with the effect of adding nano CeO2 to act as a grain pinner/refiner to a known Cu-Al-Ni shape memory alloy. Elements were taken in a predefined ratio to prepare 300 g alloy per batch and melted in an induction furnace. Casting was followed by homogenization at 1173 K (900 °C) and rolling to make sheets of 0.5-mm thickness. Furth...

View

Effect of Mn Content on the Properties Affecting Shape Memory Behaviour of Cu-12Al-4Ni-10Zn Alloy

ArticleMar 2016

Rupa Dasgupta · Ashish Kumar Jain · Shahadat Hussain · [...] · Sampath Vedamanickam

View

Effect of Varying Al/Mn Ratio on Phase Transformation in Cu–Al–Mn Shape Memory Alloys

ArticleFull-text availableNov 2015

Ashish Kumar Jain · Shahadat Hussain · Pravir Kumar · [...] · Rupa Dasgupta

In the present study, an attempt has been made to study the effect of the proportion of the main alloying constituents in a Cu–Al alloys with alloying additions of Mn, Ni and Zn in varying proportions and combinations. The alloys have been subjected to heat treatment cycles to improve upon the microstructure and precipitate the required martensite phase. After appropriate...

View

Effects of Different Quaternary Additions in the Properties of a Cu-Al-Mn Shape Memory Alloy

ArticleNov 2015

Shahadat Hussain · Pravir Kumar · Ashish Kumar Jain · [...] · Rupa Dasgupta

In the present study, Cu–12.5 wt. %Al–5 wt. %Mn–80.5 wt. % shape memory alloy is chosen as the base alloy and 2 wt. % of quaternary additions of Fe, Ni, and Ti added to the base alloy. The effects of these additions on the terms of feasibility, microstructure, hardness, and transformation temperature were studied. The findings suggest the possiblitt...

View

Changes in the properties of Cu-Al-Mn shape memory alloy due to quaternary addition of different elements

ArticleFull-text availableMar 2015

Pravir Kumar · Ashish Kumar Jain · Shahadat Hussain · [...] · Rupa Dasgupta

Advantages of Cu based shape memory alloy include amongst other features, high transformation temperature, low cost of production, ease in manufacturing processes and ability to vary the achieved properties through alloying additions. It has been often reported that these alloys are very sensitive to the alloying additions in terms of properties ac...

View

Role of alloying additions on the properties of Cu–Al–Mn shape memory alloys

ArticleJan 2015

Rupa Dasgupta · Ashish Kumar Jain · Pravir Kumar · [...] · Abhishek Pandey

View

Effect of alloying constituents on the martensitic phase formation in some Cu-based SMAs

ArticleFull-text availableJul 2014

Rupa Dasgupta · Ashish Kumar Jain · Pravir Kumar · [...] · Abhishek Pandey

The paper discusses the attempt made to improve upon the transition temperatures and amount of martensite formation in Cu–Al alloys with alloying additions of Mn, Ni and Zn in varying proportions and combinations. The alloys have been subjected to heat treatment cycles to improve upon the microstructure and precipitate the required martensite phase...

View

Questions

Questions (3)

Can anyone help to get PDF Card of Al2Mg phase of JCPDS Card no. is 34-1035?

QuestionSep 2017

XRD data analysis of samples

View

How to obtain PDF 4+ database for XRD analysis using X'pert High Score software?

QuestionAug 2016

Digital format of PDF 4+ database which is issued by ICDD.

View

What are 2H and 18R martensites? Can anyone explain these to me?

QuestionJan 2016

With respect to Shape memory alloys

View

5 Answers

Network

Cited

View All

Christoph Haberland

Siemens AG, Power and Gas Division

Vamsi Krishna Balla

Sanjay Kumar Vajpai

National Institute of Technology Jamshedpur

I. Karaman

Texas A&M University

Osman Adiguzel

Firat University

Cited By

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Wael Zaki

Khalifa University

Mediha Kök

Firat University

Chunze Yan

Huazhong University of Science and Technology

Palani Anand iyamperumal

Indian Institute of Technology Indore

Tamara Holjevac Grgrurić

Faculty of metallurgy University of Zagreb

